

Drive Test Report

Mumbai , India

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Table of Contents

1. Introduction	3
2. Area Summary	3
3. Drive Summary	4
4. Coverage KPI Analysis	6

1. Introduction

This telecom drive test report provides an overview of the performance of wireless communication services in Mumbai, India. Drive testing is a critical process that assesses signal strength, coverage, and overall provider performance, enabling service providers to identify areas of strong and weak coverage. This report presents map visualizations highlighting the distribution of coverage areas, providing valuable insights into network performance. The findings presented in this report offer actionable recommendations for optimization and deployment planning, supporting informed decision-making to enhance the quality of wireless services.

2. Area Summary

Drive route covers key operational areas identified based on spatial distribution of samples and session density.

Hotspots & Marked Locations

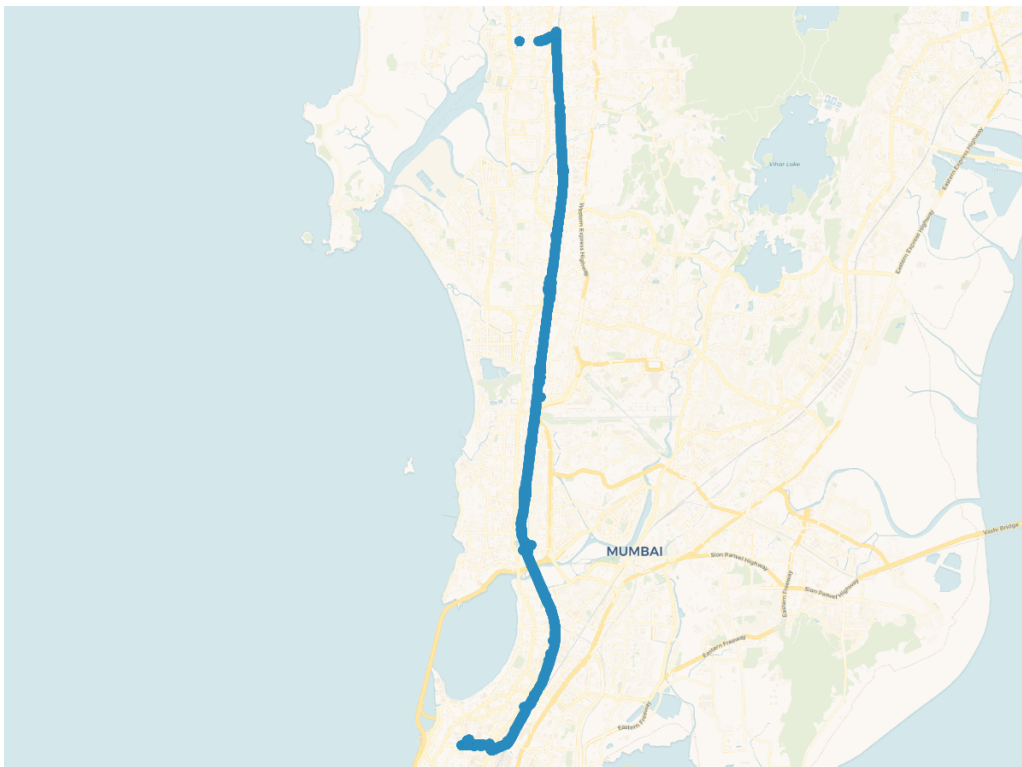
Hanuman Nagar, Balasheth Mandurkar Marg, Rai Pada

Crowded & High-Traffic Locations

Rai Pada

Major Areas Covered

The drive covered major areas including Hanuman Nagar (highway residential, 15811 samples), Balasheth Mandurkar Marg (highway residential, 835 samples), Rai Pada (highway residential, 633 samples), Kasturba Road (highway residential, 284 samples), Pandurang Budhkar Marg (highway secondary, 233 samples), Dadar West (highway primary, 171 samples).



3. Drive Summary

The drive test was conducted over 3 days from 2026-05-22 to 2026-06-05, covering 36.9 km with 23,306 samples collected across 3 sessions.

Metric	Value
Distance Covered	36.9 km
Total Samples	23,306
Total Sessions	3
Number of Days	3 Days

Session Details

Session	Date	Start	End	Duration	Distance
4530	22-05-2026	14:11	14:22	11 min	0.66 km
4714	04-06-2026	13:38	18:23	4 hours 45 min	20.57 km
4742	05-06-2026	07:14	08:10	55 min	15.62 km

Executive Summary

Objective

The objective of this drive test was to evaluate LTE radio coverage, radio quality, mobility performance and end-user service experience along the planned drive route. Measurements were collected using STracer and analysed against operator acceptance criteria.

Scope

- Coverage assessment using Band Distribution, RSRP, RSRQ and SINR.
- Mobility assessment using Serving Cell, PCI Distribution and Neighbor behaviour.
- Handover verification including Intra-frequency, Inter-frequency and Inter-RAT events where available.
- Service quality evaluation using Throughput, MOS, Latency, Jitter and Packet Loss.

Executive KPI Summary

KPI Group	Status	Remarks
Coverage	Good	Average RSRP -71.9 dBm; 94% of samples above -95 dBm.
Radio Quality - 4G (LTE Anchor - NSA)	Fair	[13,574 samples] Average SINR 19.5 dB; Average RSRQ -7.3 dB.
Radio Quality - 4G	Fair	[5,909 samples] Average SINR 18.7 dB; Average RSRQ -7.2 dB.
Radio Quality - 5G NSA	Fair	[3,823 samples] Average SINR 23.8 dB; Average RSRQ -10.3 dB.
Mobility	Good	129 unique serving PCIs (top PCI 41% of samples).
Handover	Good	891 band handover events; no major mobility interruption observed.

Key Engineering Observations

- Coverage classified as Good (94% of RSRP samples above -95 dBm, average -71.9 dBm).
- Radio quality (4G (LTE Anchor - NSA)) classified as Fair. Requires optimization.
- Radio quality (4G) classified as Fair. Requires optimization.
- Radio quality (5G NSA) classified as Fair. Requires optimization.

Overall Assessment

The network showed mixed RF performance with some KPI groups below the Good threshold. Targeted optimization is recommended for the KPI groups flagged Fair or Poor above.

4. Coverage KPI Analysis

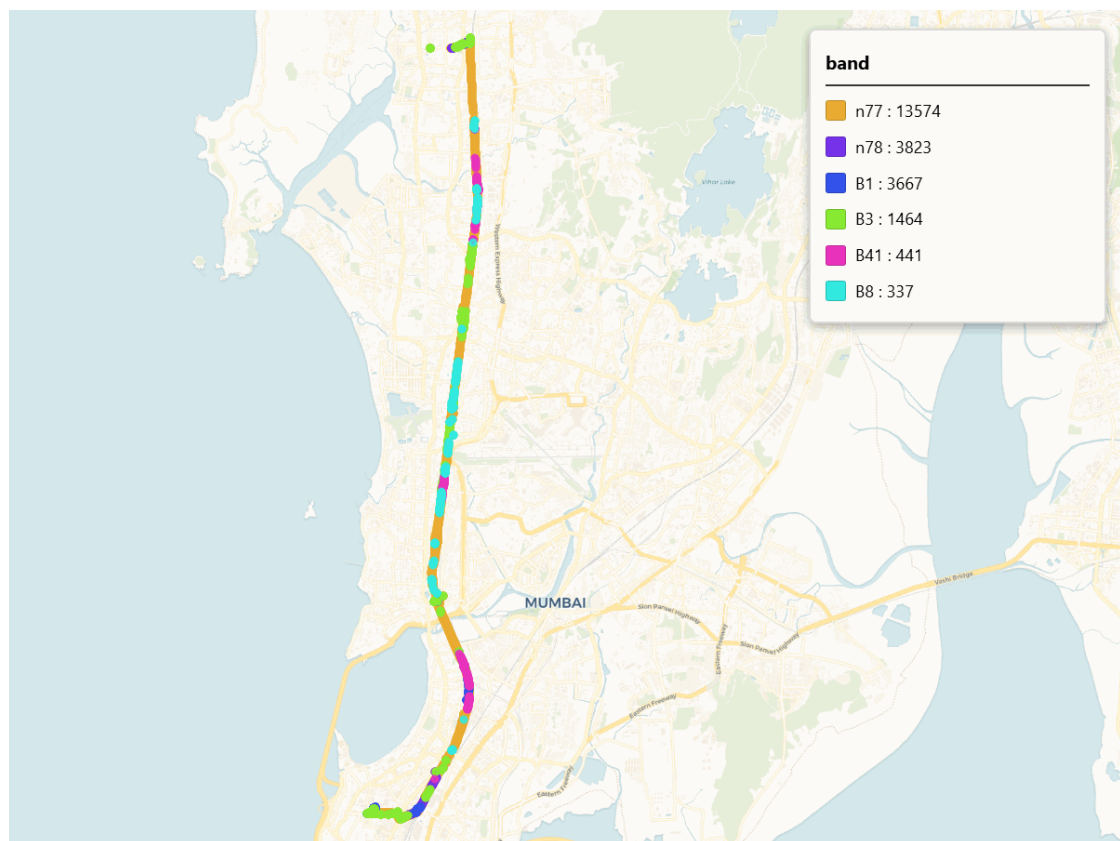
Coverage analysis is the first step in RF optimization. It evaluates whether the radio network provides continuous service with acceptable signal strength and quality across the planned drive route.

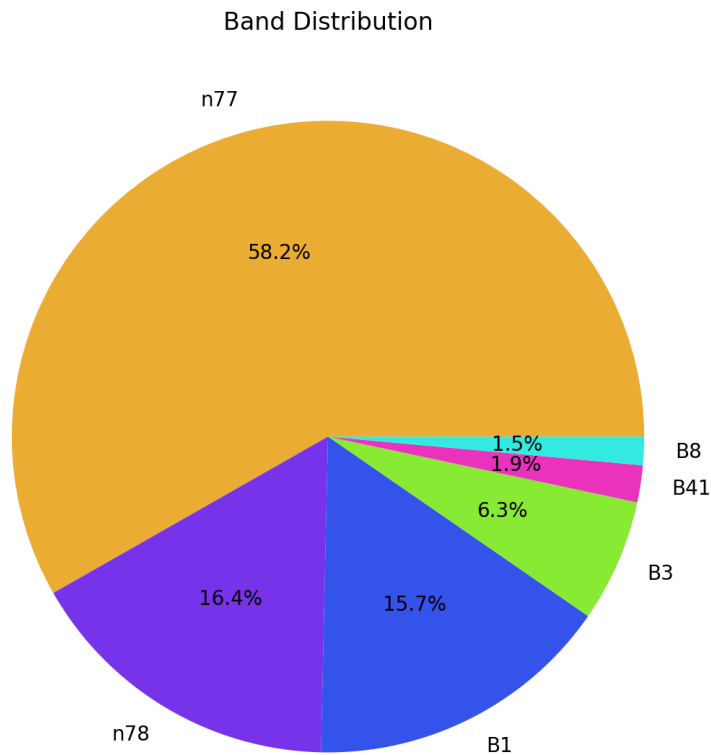
4.1 Band Distribution

Objective: verify that the UE camps on the planned band(s) throughout the drive route and identify unexpected band changes.

Band	Sample Count	% Samples
n77	13,574	58.2%
n78	3,823	16.4%
B1	3,667	15.7%
B3	1,464	6.3%
B41	441	1.9%
B8	337	1.4%

Observation: The majority of samples were served on Band n77 (58.2%), with additional coverage on Band n78 (16.4%), Band B1 (15.7%). No unexpected frequency changes were observed.





4.2 RSRP Analysis (Coverage)

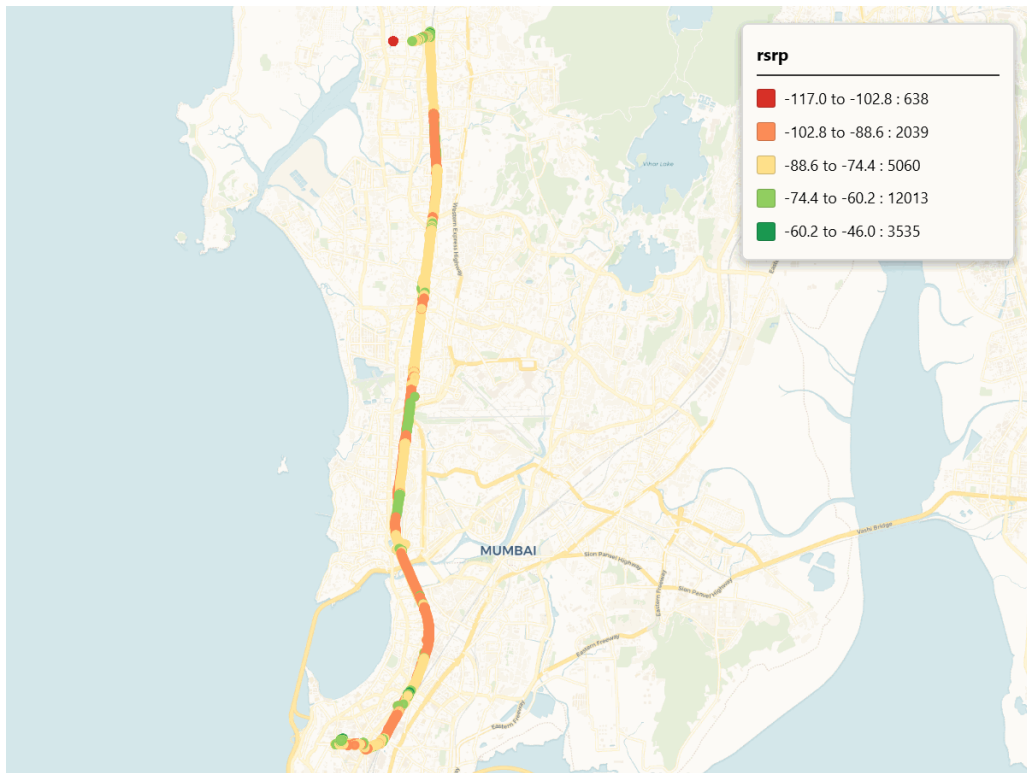
Definition: Reference Signal Received Power (RSRP) is the primary LTE coverage KPI and indicates the received signal strength from the serving cell.

Acceptance Criteria

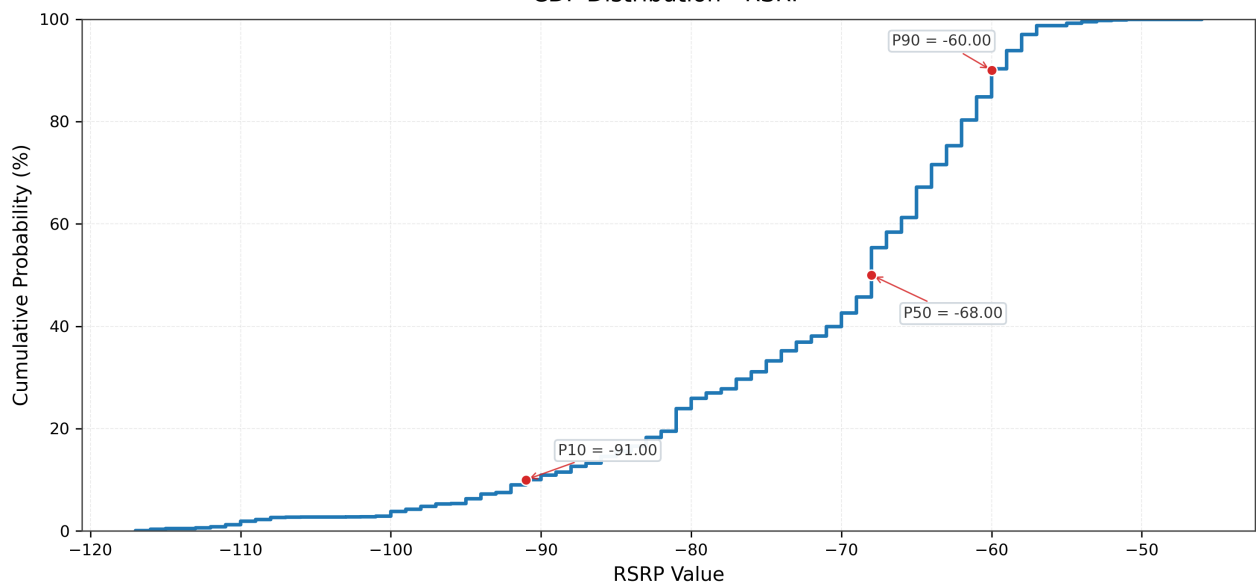
- Good: RSRP > -95 dBm for more than 90% of samples
- Fair: Average RSRP -95 to -105 dBm
- Poor: Average RSRP < -105 dBm, or coverage (% > -95 dBm) < 75%

Metric	Sample Value
Average RSRP	-71.9 dBm
Best RSRP	-46.0 dBm
Worst RSRP	-117.0 dBm
Samples > -95 dBm	94%

Observation: Coverage classified as **Good**. Average RSRP -71.9 dBm; 94% of samples above -95 dBm.



CDF Distribution - RSRP



4.3 RSRQ Analysis

Definition: Reference Signal Received Quality (RSRQ) indicates the quality of the received reference signal and reflects both signal strength and cell loading. RSRQ is classified PER TECHNOLOGY, not blended, since its characteristic range differs by RAT.

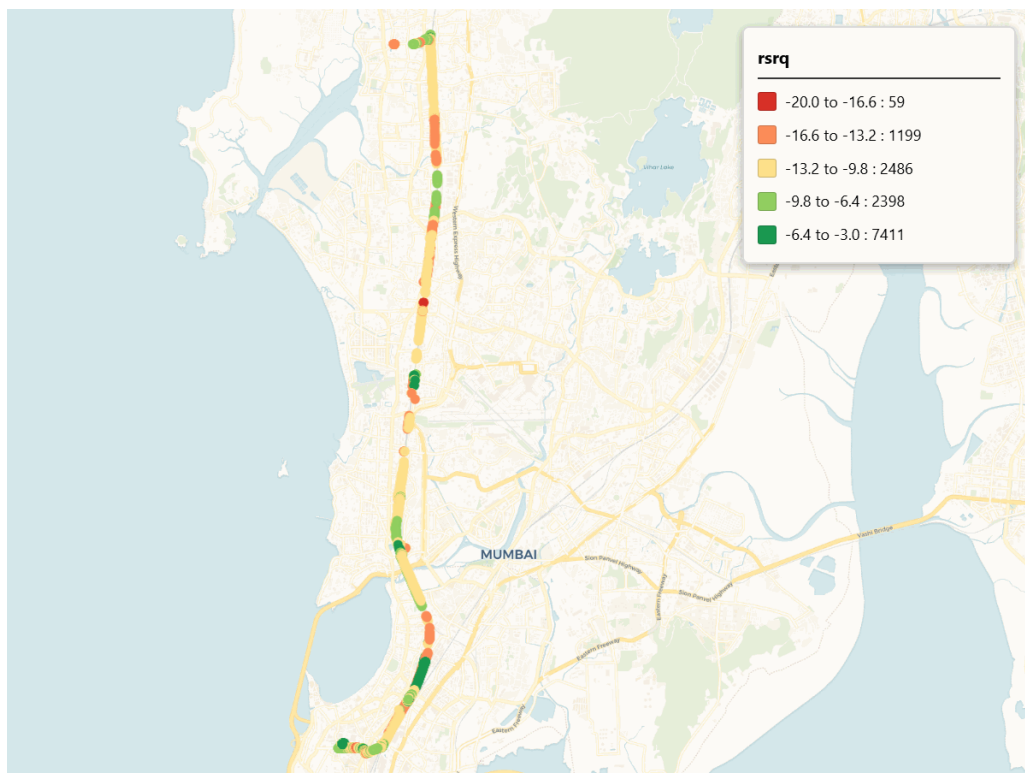
Acceptance Criteria

- Good: Average RSRQ > -10 dB
- Fair: Average RSRQ -10 to -15 dB
- Poor: Average RSRQ < -15 dB

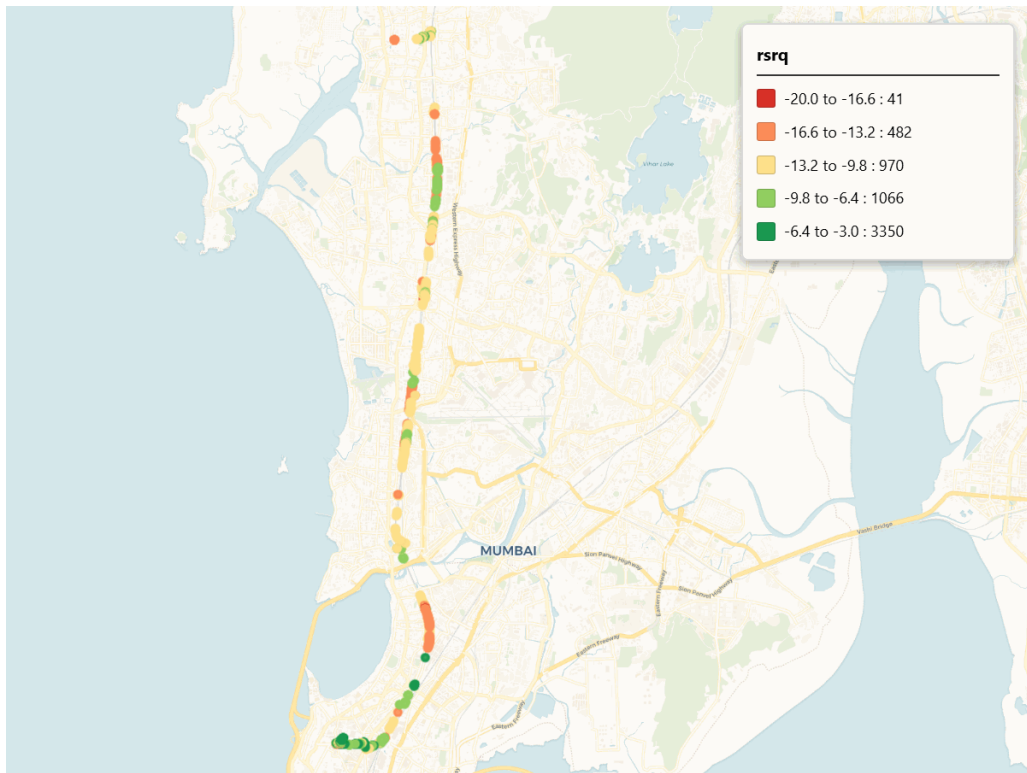
Technology	Average RSRQ	Worst RSRQ	Status
4G (LTE Anchor - NSA)	-7.3 dB	-20.0 dB	Good
4G	-7.2 dB	-20.0 dB	Good
5G NSA	-10.3 dB	-15.0 dB	Fair

Observation: worst-case RSRQ classification across technologies is **Fair**.

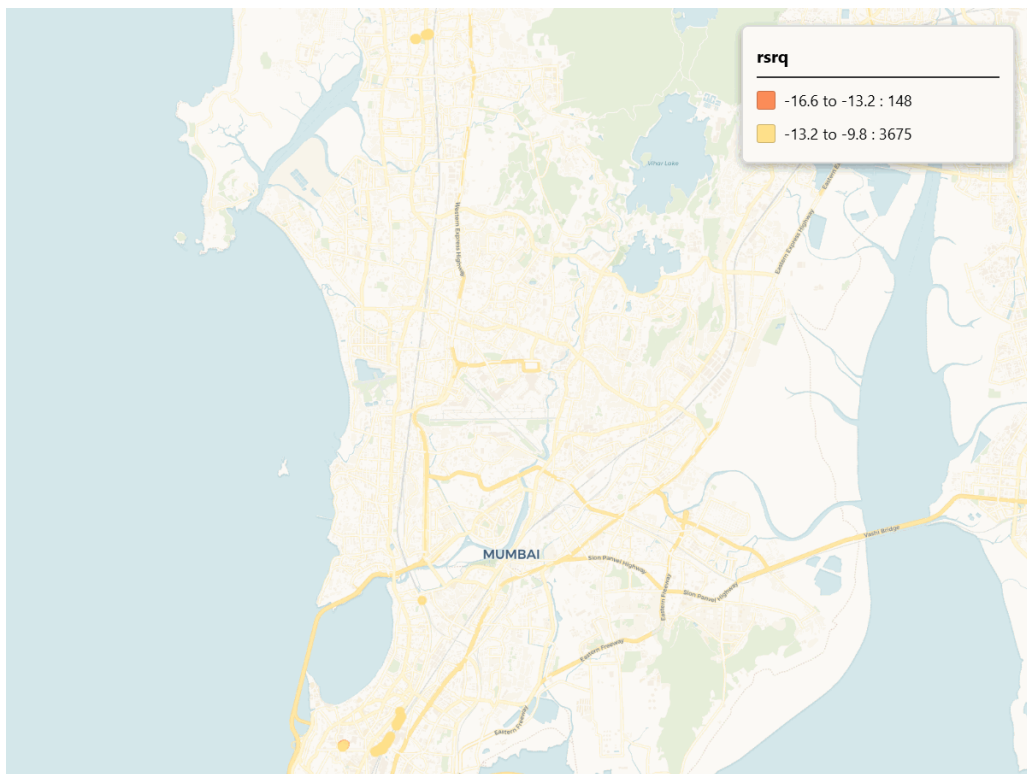
4G (LTE Anchor - NSA)

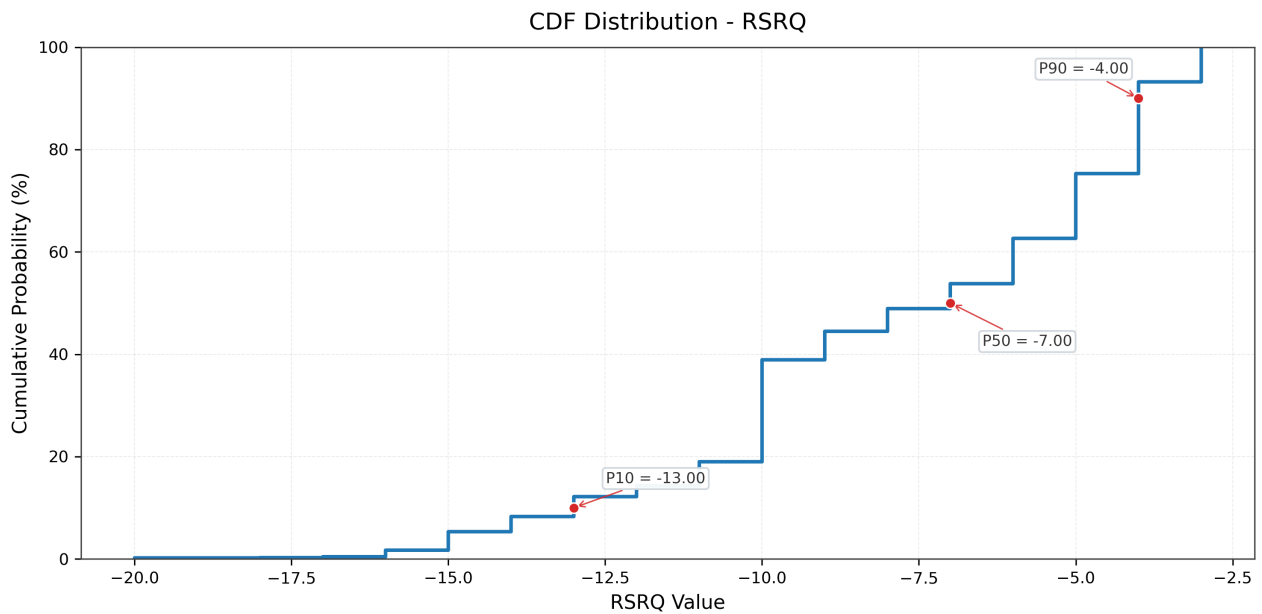


4G



5G NSA





4.4 SINR Analysis

Definition: Signal-to-Interference-plus-Noise Ratio (SINR) represents radio link quality and interference conditions. SINR is classified PER TECHNOLOGY, not blended, since its characteristic range differs by RAT.

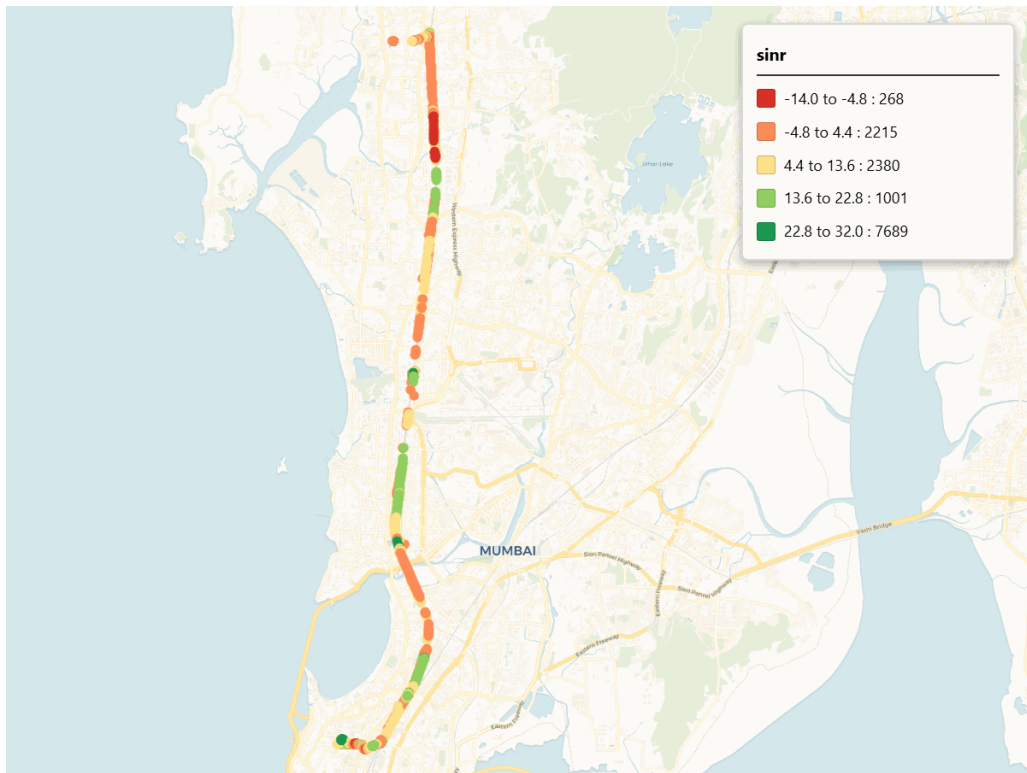
Acceptance Criteria

- Good: Average SINR > 20 dB
- Fair: Average SINR 13 to 20 dB
- Poor: Average SINR < 13 dB

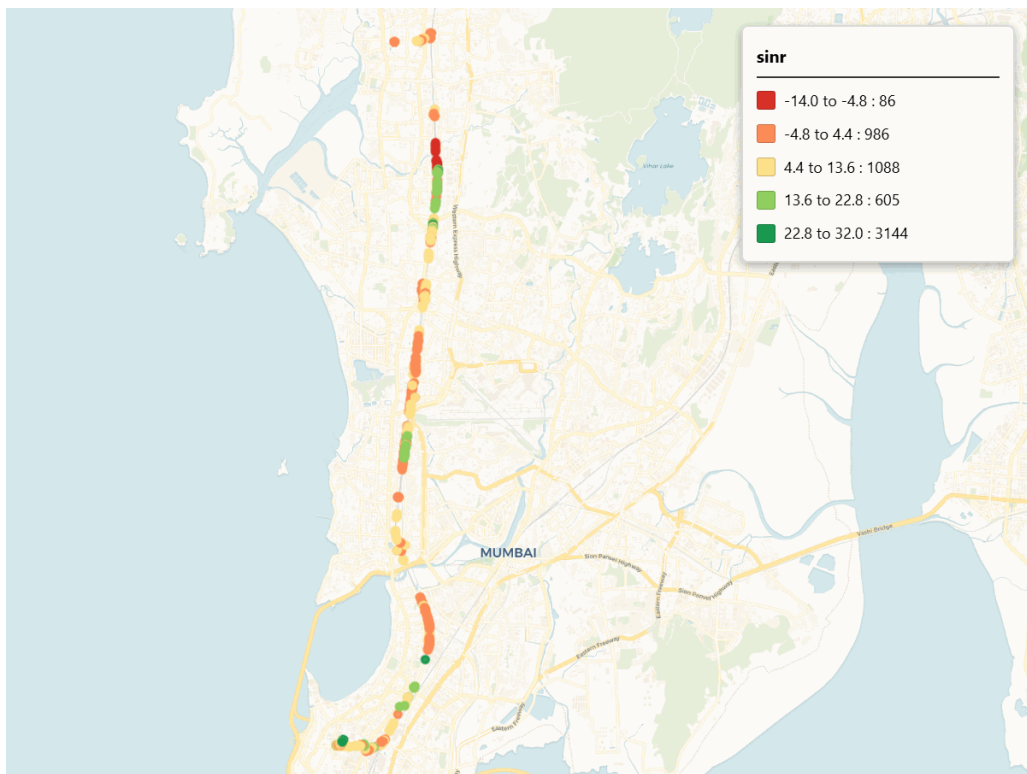
Technology	Average SINR	Worst SINR	Status
4G (LTE Anchor - NSA)	19.5 dB	-14.0 dB	Fair
4G	18.7 dB	-14.0 dB	Fair
5G NSA	23.8 dB	-4.0 dB	Good

Observation: worst-case SINR classification across technologies is **Fair**.

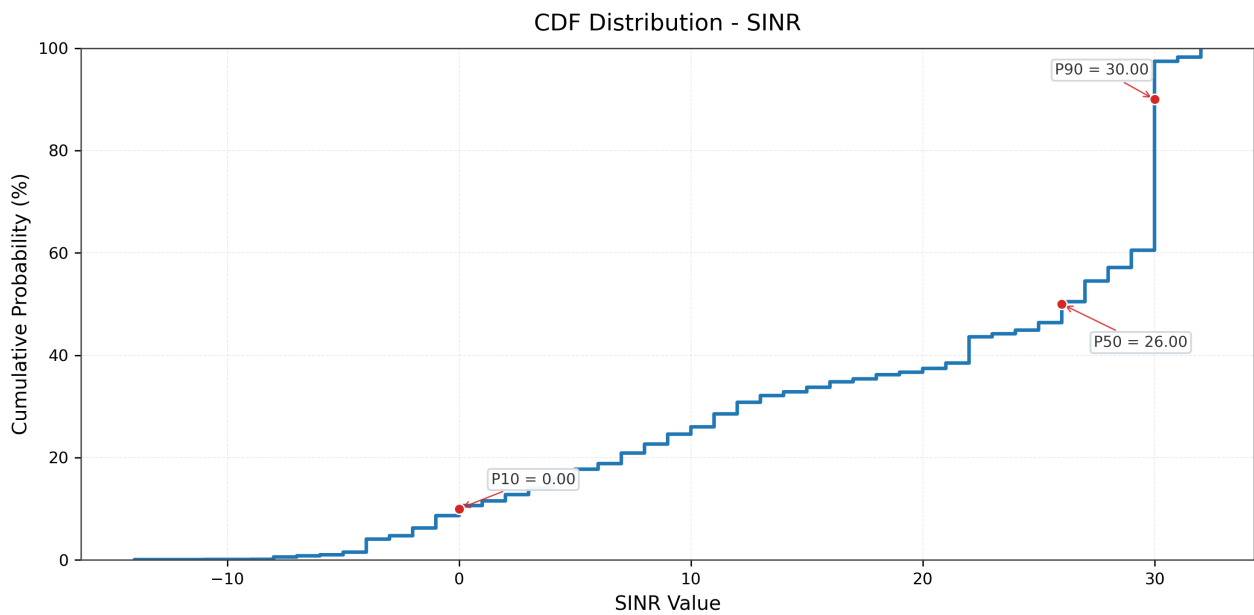
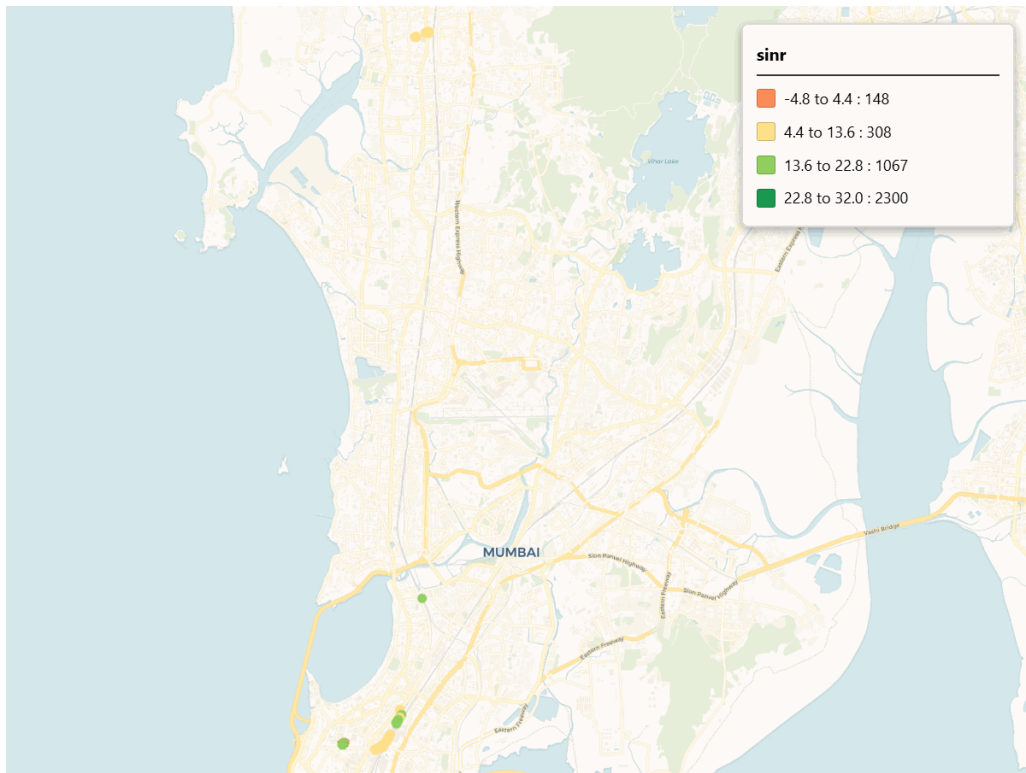
4G (LTE Anchor - NSA)



4G



5G NSA



4.5 Coverage KPI Summary

KPI	Acceptance	Observed	Status	% Within Acceptance	% Outside Acceptance
RSRP	> -95 dBm (90% of samples)	-71.9 dBm	PASS	94%	6%
RSRQ - 4G (LTE Anchor - NSA)	> -10 dB	-7.3 dB	PASS	72%	28%
RSRQ - 4G	> -10 dB	-7.2 dB	PASS	75%	25%
RSRQ - 5G NSA	> -10 dB	-10.3 dB	FAIL	0%	100%
SINR - 4G (LTE Anchor - NSA)	> 20 dB	19.5 dB	FAIL	59%	41%
SINR - 4G	> 20 dB	18.7 dB	FAIL	56%	44%
SINR - 5G NSA	> 20 dB	23.8 dB	PASS	87%	13%

Overall Coverage Assessment

The coverage KPIs indicate localized weak-signal or quality pockets. Targeted optimization is recommended for the specific KPI/technology combinations flagged Fair or Poor above.